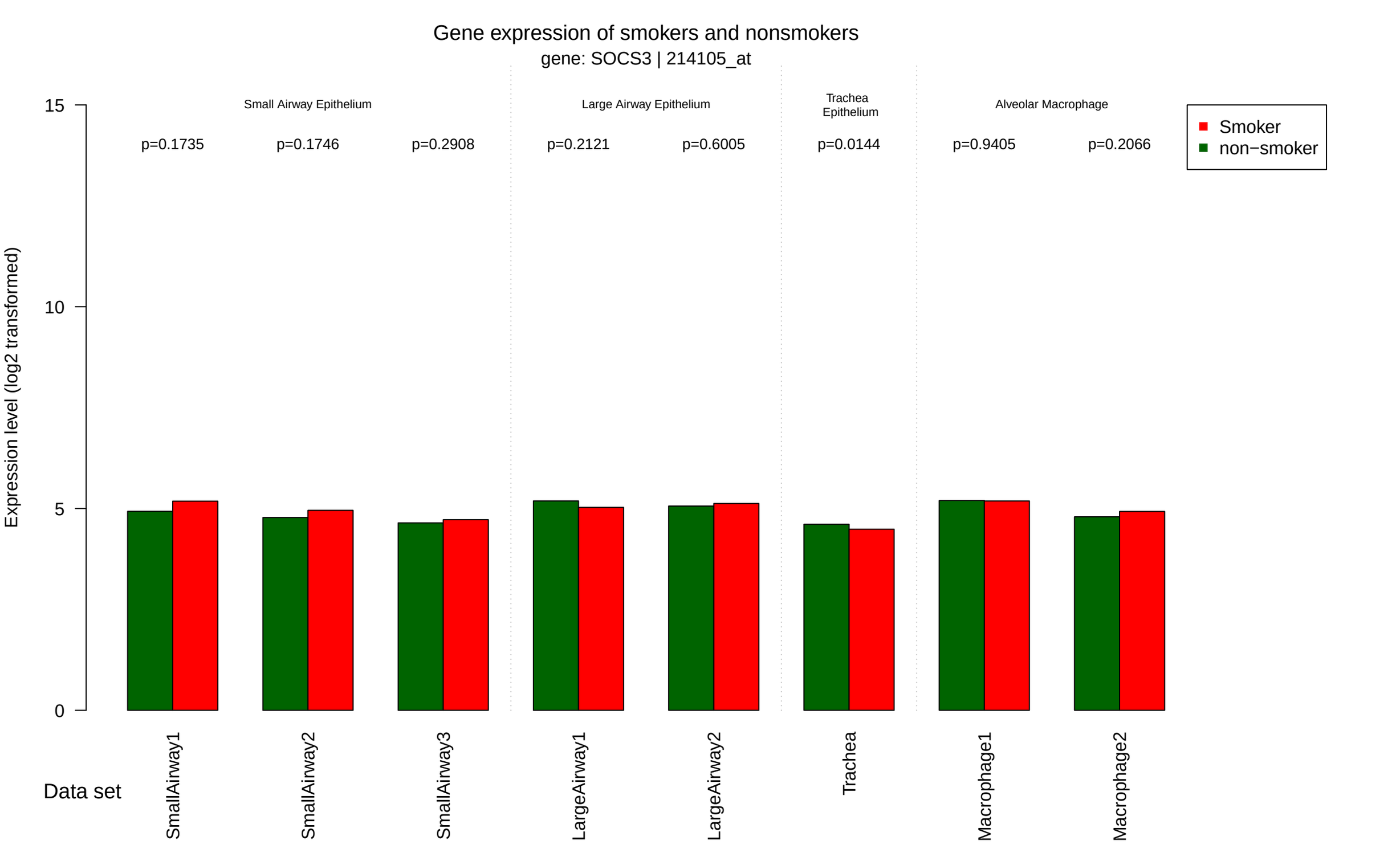
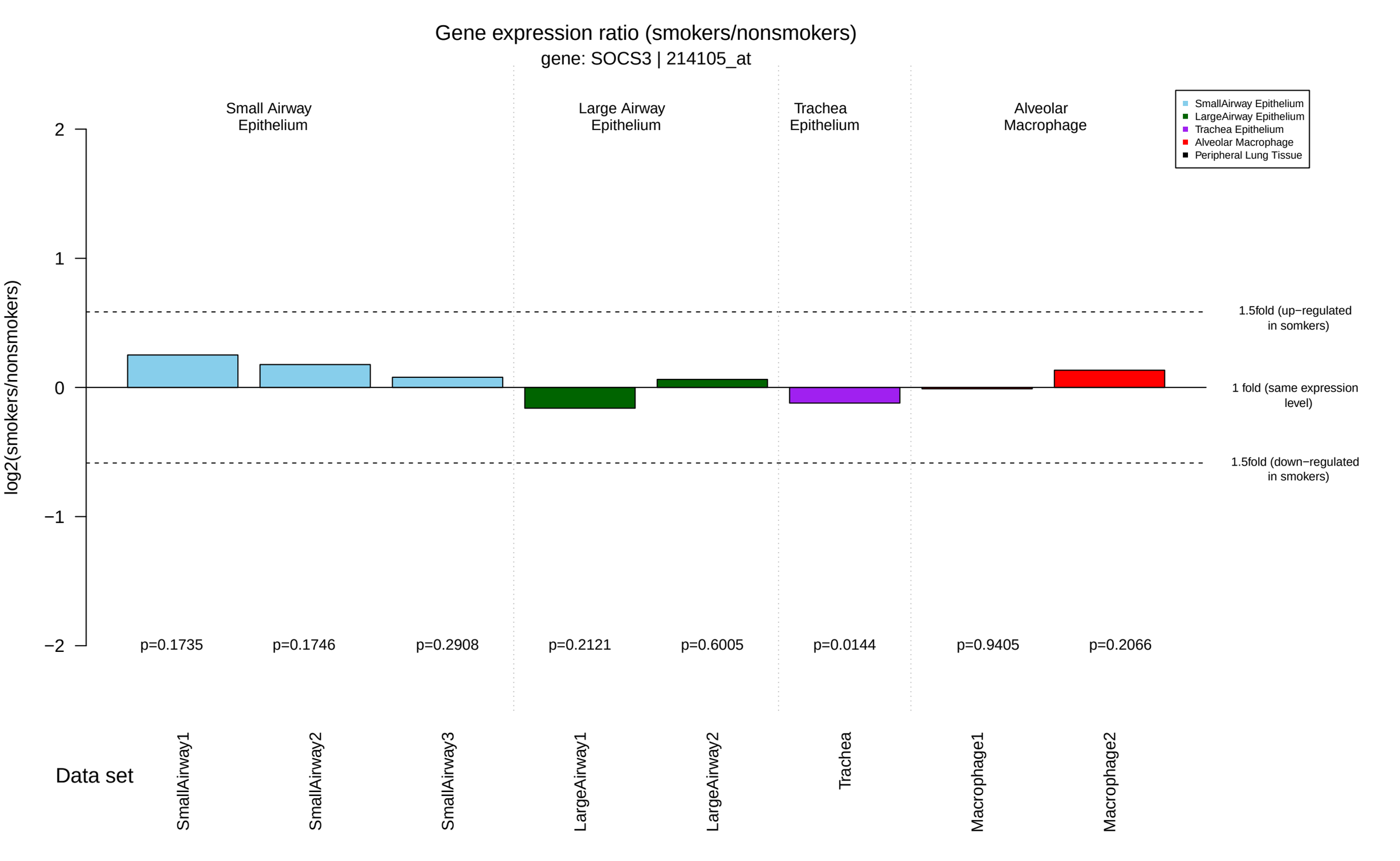
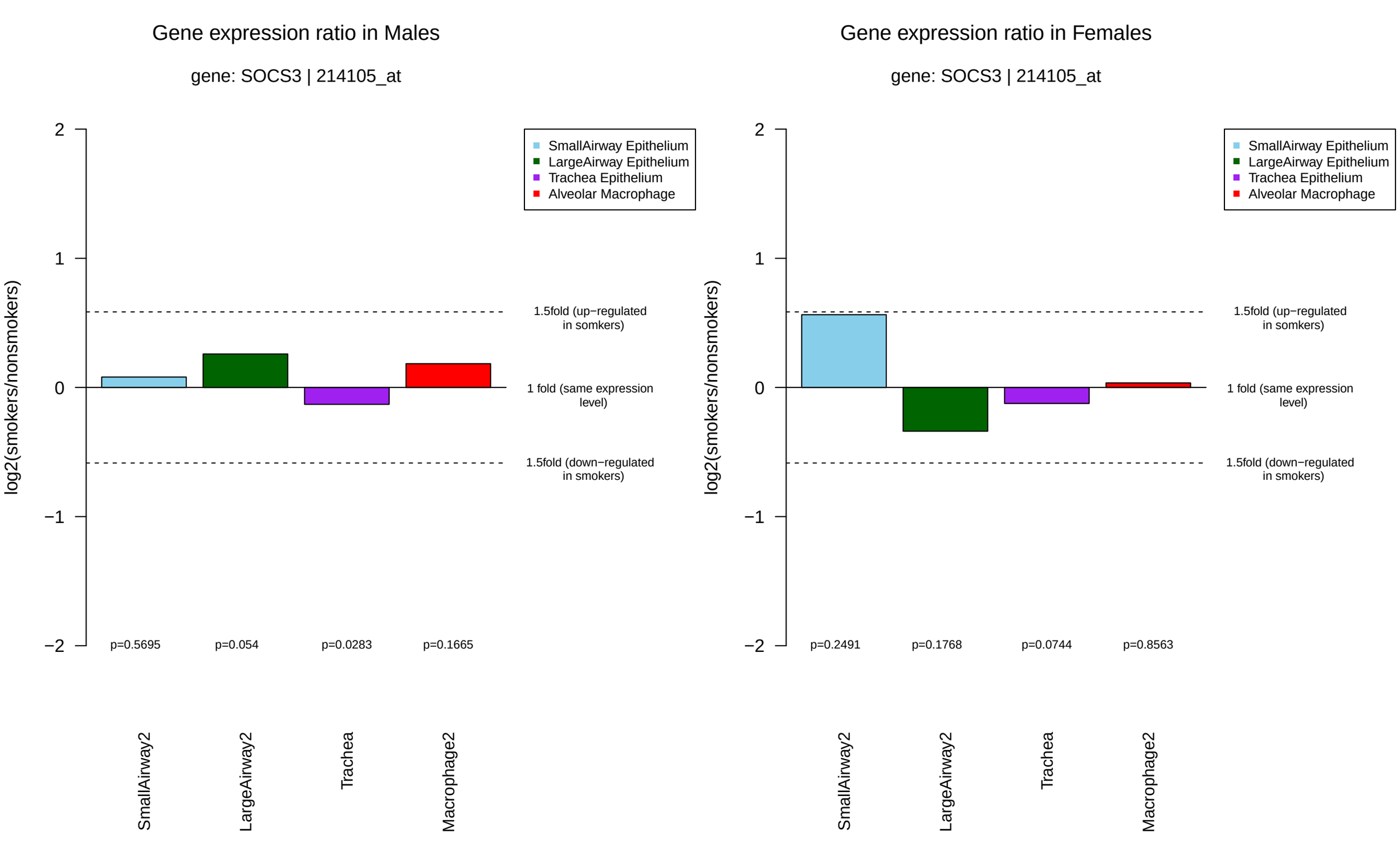
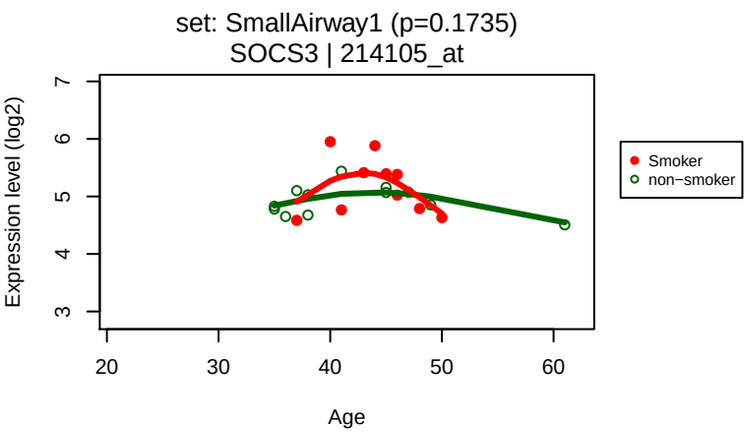
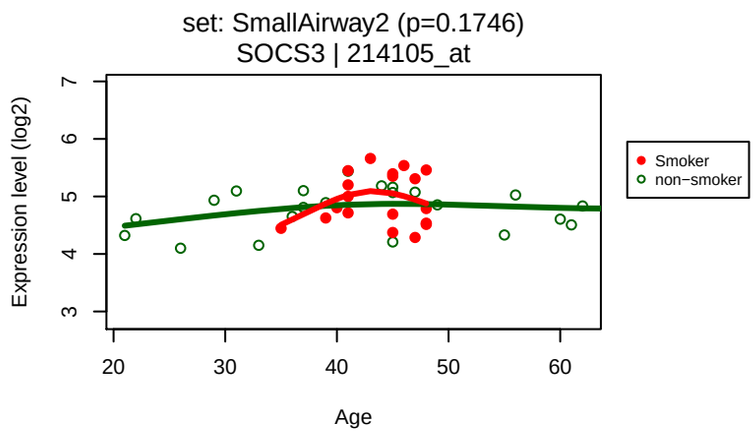
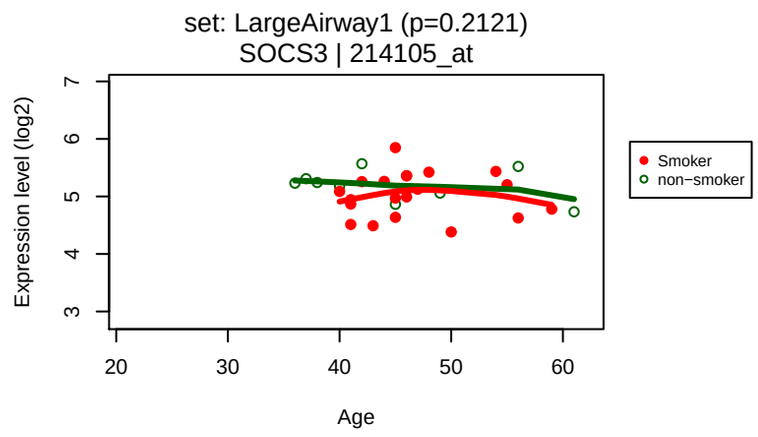

Gene expression of smokers and nonsmokers
gene: SOCS3 | 214105_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1735
p=0.1746
p=0.2908
p=0.2121
p=0.6005
p=0.0144
p=0.9405
p=0.2066
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

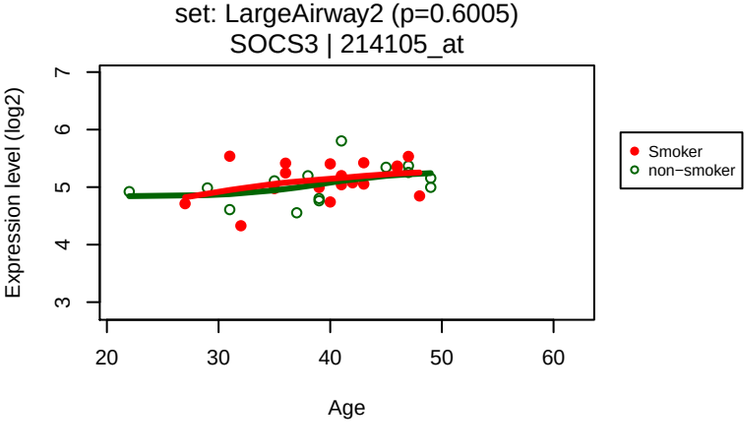

Gene expression ratio (smokers/nonsmokers)
gene: SOCS3 | 214105_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1735
p=0.1746
p=0.2908
p=0.2121
p=0.6005
p=0.0144
p=0.9405
p=0.2066
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

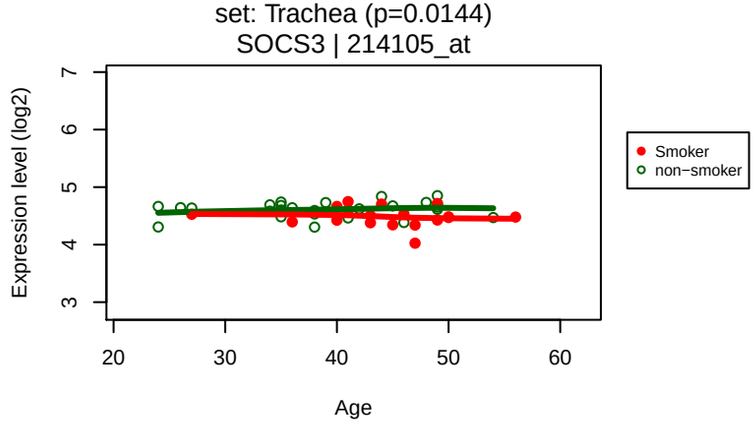

Gene expression ratio in Males
gene: SOCS3 | 214105_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5695
p=0.054
p=0.0283
p=0.1665
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOCS3 | 214105_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2491
p=0.1768
p=0.0744
p=0.8563
SmallAirway2
LargeAirway2
Trachea
Macrophage2

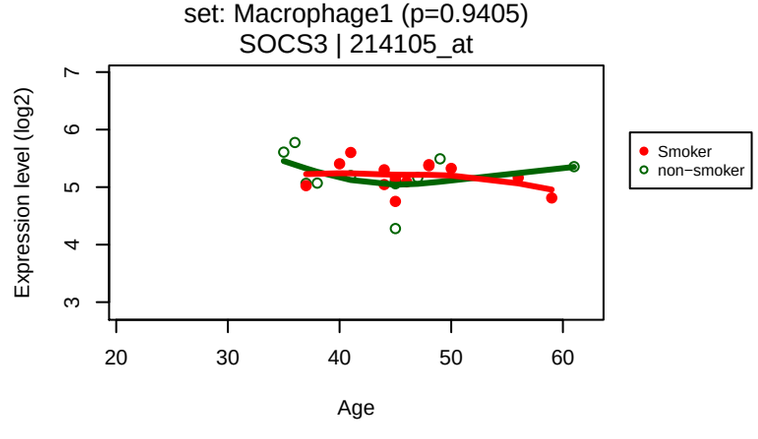

set: SmallAirway1 (p=0.1735)
SOCS3 | 214105_at
Expression level (log2)
Age
Smoker
non−smoker

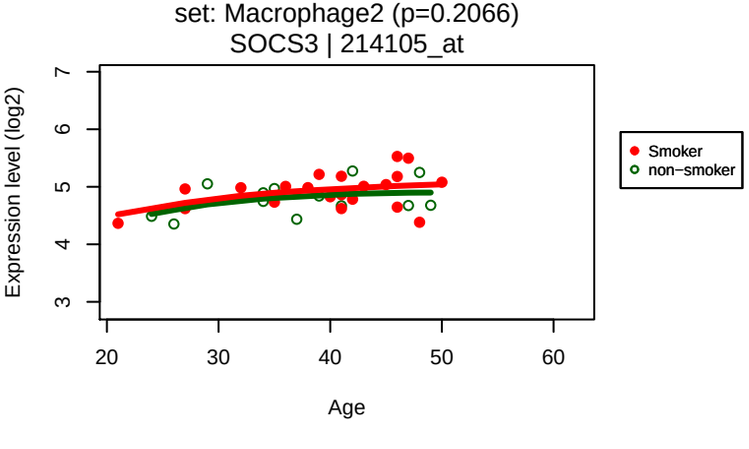

set: SmallAirway2 (p=0.1746)
SOCS3 | 214105_at
Expression level (log2)
Age
Smoker
non−smoker

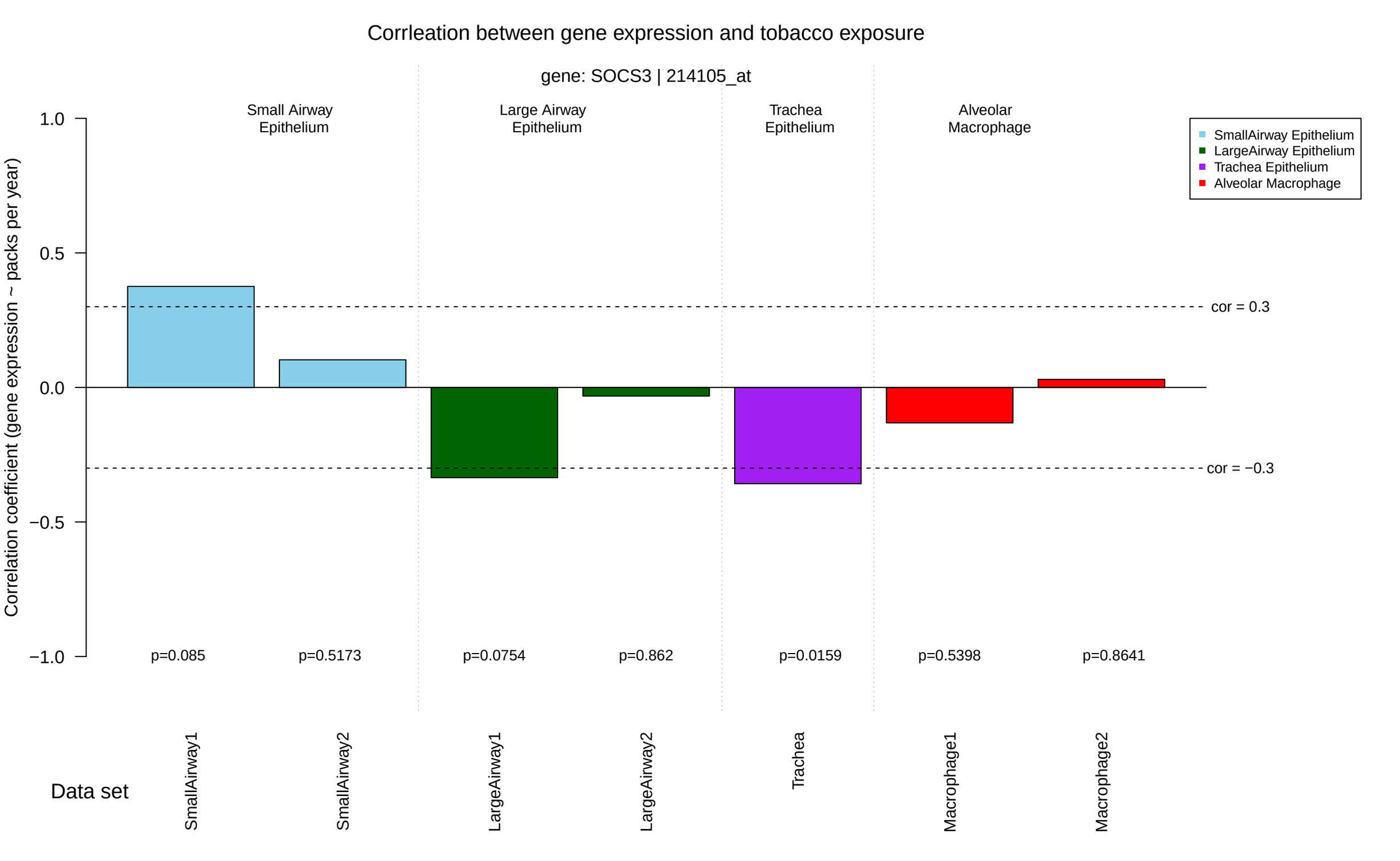

set: LargeAirway1 (p=0.2121)
SOCS3 | 214105_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6005)
SOCS3 | 214105_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0144)
SOCS3 | 214105_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9405)
SOCS3 | 214105_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2066)
SOCS3 | 214105_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOCS3 | 214105_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.085
p=0.5173
p=0.0754
p=0.862
p=0.0159
p=0.5398
p=0.8641
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2